

Cloud Shell

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https://shell.cloud.google.com/?pli=1&show=ide%2Cterminal

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Cloud Shell Editor

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```
pyogesh2424@cloudshell:~$ uname
Linux
pyogesh2424@cloudshell:~$
pyogesh2424@cloudshell:~$
pyogesh2424@cloudshell:~$
Welcome to Cloud Shell! Type "help" to get started, or type "gemini" to try prompting with Gemini CLI.
To set your Cloud Platform project in this session use `gcloud config set project [PROJECT_ID]`.
You can view your projects by running `gcloud projects list`.
pyogesh2424@cloudshell:~$ uname
Linux
pyogesh2424@cloudshell:~$
pyogesh2424@cloudshell:~$
pyogesh2424@cloudshell:~$
```

```
top - 09:14:15 up 34 min,  0 user,  load average: 0.02, 0.63, 0.64
Tasks:  32 total,   1 running,  31 sleeping,   0 stopped,   0 zombie
%Cpu(s):  4.5 us,  4.5 sy,   0.0 ni, 90.9 id,   0.0 wa,   0.0 hi,   0.0 si,   0.0 st
MiB Mem :  7947.4 total,  4093.7 free,  2202.4 used,  1897.3 buff/cache
MiB Swap:   0.0 total,   0.0 free,   0.0 used.  5744.9 avail Mem

  PID USER      PR  NI    VIRT    RES    SHR S  %CPU  %MEM    TIME+  COMMAND
    1 root        20   0   4324    3268   2984 S   0.0   0.0   0:00.07  bash
    9 syslog     20   0  222140    3724   2372 S   0.0   0.0   0:00.02  rsyslogd
   25 root        20   0   38768   27788  10684 S   0.0   0.3   0:00.80  python
   26 root        20   0    6196    2680   2440 S   0.0   0.0   0:00.00  logger
   68 root        10 -10   12024    4276    3160 S   0.0   0.1   0:00.00  sshd
top - 09:14:18 up 34 min,  0 user,  load average: 0.02, 0.63, 0.64
 448 root        20   0 1234268   13096   8276 S   0.0   0.2   0:00.48  editor-+
%Cpu(s):  0.5 us,  0.3 sy,   0.0 ni, 99.0 id,   0.0 wa,   0.0 hi,   0.0 si,   0.2 st
Help for Interactive Commands - procps-ng 4.0.4
Window 1:Def: Cumulative mode Off.  System: Delay 3.0 secs; Secure mode Off.

Z,B,E,e  Global: 'Z' colors; 'B' bold; 'E'/'e' summary/task memory scale
l,t,m,I,0 Toggle: 'l' load avg; 't' task/cpu; 'm' memory; 'I' Irix; '0' zeros
```

Cloud Shell

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by selecting a number from: 1 =Def; 2 =Job; 3 =Mem; or 4 =Usr

. Commands requiring 'A' mode -----

G . Change the Name of the 'current' window/field group

* a , w . Cycle through all four windows: 'a' Forward; 'w' Backward

* - , _ . Show/Hide: '-' Current window; '_' all Visible/Invisible

The screen will be divided evenly between task displays. But you can make some larger or smaller, using 'n' and 'i' commands. Then later you could:

* = , + . Rebalance tasks: '=' Current window; '+' Every window

(this also forces the current or every window to become visible)

In 'A' mode, '[' keys are your essential commands. Please try the 'a' and 'w' commands plus the 'g' sub-commands NOW. Press <Enter> to make 'Current'

Help for Windows / Field Groups - "Current Window" = 1:Def

. Use multiple windows, each with separate config opts (color,fields,sort,etc)

. The 'current' window controls the Summary Area and responds to your Commands

. that window's task display can be turned Off & On, growing/shrinking others

. with NO task display, some commands will be disabled ('i','R','n','c', etc)

. until a different window has been activated, making it the 'current' window

. You change the 'current' window by: 1) cycling forward/backward; 2) choosing a specific field group; or 3) exiting the color mapping or fields screens

. Commands available anytime -----

A . Alternate display mode toggle, show Single / Multiple windows

g . Choose another field group and make it 'current', or change now

top - 09:14:56 up 34 min, 0 user, load average: 0.11, 0.58, 0.63

Tasks: 32 total, 1 running, 31 sleeping, 0 stopped, 0 zombie

%Cpu(s): 2.2 us, 1.7 sy, 0.0 ni, 96.1 id, 0.0 wa, 0.0 hi, 0.0 si, 0.0 st

MiB Mem : 7947.4 total, 4092.0 free, 2204.1 used, 1897.4 buff/cache

MiB Swap: 0.0 total, 0.0 free, 0.0 used. 5743.3 avail Mem

PID	USER	PR	NI	VIRT	RES	SHR	S	%CPU	%MEM	TIME+	COMMAND
460	root	20	0	1793224	45892	32276	S	0.3	0.6	0:01.87	containerd
1574	pyogesh+	20	0	1016648	61212	46100	S	0.3	0.8	0:00.32	node
1	root	20	0	4324	3268	2984	S	0.0	0.0	0:00.07	bash
9	syslog	20	0	222140	3724	2372	S	0.0	0.0	0:00.02	rsyslogd
25	root	20	0	38768	27788	10684	S	0.0	0.3	0:00.80	python

pyogesh2424@cloudshell:~\$ htop

pyogesh2424@cloudshell:~\$ lsb_release

No LSB modules are available.

pyogesh2424@cloudshell:~\$ arch

x86_64

pyogesh2424@cloudshell:~\$ uname

Linux

pyogesh2424@cloudshell:~\$

```
Cloud Shell
https://shell.cloud.google.com/?pli=1&show=ide%2Cterminal
Cloud Shell Editor
cloudshell X +
pyogesh2424@cloudshell:~$ uname
Linux
pyogesh2424@cloudshell:~$ mkdir linux_practice
pyogesh2424@cloudshell:~$ cd linux_practice
pyogesh2424@cloudshell:~/linux_practice$ pwd
/home/pyogesh2424/linux_practice
pyogesh2424@cloudshell:~/linux_practice$ ls
pyogesh2424@cloudshell:~/linux_practice$ cd
pyogesh2424@cloudshell:~$ touch file1.txt
touch file2.txt
pyogesh2424@cloudshell:~$ ls
file1.txt file2.txt linux_practice README-cloudshell.txt
pyogesh2424@cloudshell:~$ mkdir projects
pyogesh2424@cloudshell:~$ cd projects
pyogesh2424@cloudshell:~/projects$ pwd
/home/pyogesh2424/projects
pyogesh2424@cloudshell:~/projects$ cd ..
pyogesh2424@cloudshell:~$ pwd
/home/pyogesh2424
pyogesh2424@cloudshell:~$ ^C
pyogesh2424@cloudshell:~$ mkdir notes
pyogesh2424@cloudshell:~$ ls
file1.txt file2.txt linux_practice notes projects README-cloudshell.txt
pyogesh2424@cloudshell:~$ rmdir notes
pyogesh2424@cloudshell:~$ ls
file1.txt file2.txt linux_practice projects README-cloudshell.txt
pyogesh2424@cloudshell:~$ touch data.txt
pyogesh2424@cloudshell:~$ ls
data.txt file1.txt file2.txt linux_practice projects README-cloudshell.txt
pyogesh2424@cloudshell:~$ rm file1.txt
pyogesh2424@cloudshell:~$ ls
data.txt file2.txt linux_practice projects README-cloudshell.txt
pyogesh2424@cloudshell:~$ mkdir test
cd test
touch a.txt
cd ..
pyogesh2424@cloudshell:~$ rm -r test
pyogesh2424@cloudshell:~$ cp data.txt copy.txt
pyogesh2424@cloudshell:~$ ls
copy.txt data.txt file2.txt linux_practice projects README-cloudshell.txt
pyogesh2424@cloudshell:~$ cp -r projects projects_backup
pyogesh2424@cloudshell:~$ ls
copy.txt file2.txt projects README-cloudshell.txt
data.txt linux_practice projects_backup
```



```
pyogesh2424@cloudshell:~$ rmdir notes
pyogesh2424@cloudshell:~$ ls
file1.txt  file2.txt  linux_practice  projects  README-cloudshell.txt
pyogesh2424@cloudshell:~$ touch data.txt
pyogesh2424@cloudshell:~$ ls
data.txt  file1.txt  file2.txt  linux_practice  projects  README-cloudshell.txt
pyogesh2424@cloudshell:~$ rm file1.txt
pyogesh2424@cloudshell:~$ ls
data.txt  file2.txt  linux_practice  projects  README-cloudshell.txt
pyogesh2424@cloudshell:~$ mkdir test
cd test
touch a.txt
cd ..
pyogesh2424@cloudshell:~$ rm -r test
pyogesh2424@cloudshell:~$ cp data.txt copy.txt
pyogesh2424@cloudshell:~$ ls
copy.txt  data.txt  file2.txt  linux_practice  projects  README-cloudshell.txt
pyogesh2424@cloudshell:~$ cp -r projects projects_backup
pyogesh2424@cloudshell:~$ ls
copy.txt  file2.txt  projects  README-cloudshell.txt
data.txt  linux_practice  projects_backup
pyogesh2424@cloudshell:~$ mv copy.txt renamed.txt
pyogesh2424@cloudshell:~$ ls
data.txt  linux_practice  projects_backup  renamed.txt
file2.txt  projects  README-cloudshell.txt
pyogesh2424@cloudshell:~$ mv renamed.txt projects/
pyogesh2424@cloudshell:~$ ls projects
renamed.txt
pyogesh2424@cloudshell:~$ ln data.txt hardlink.txt
pyogesh2424@cloudshell:~$ ls
data.txt  hardlink.txt  projects  README-cloudshell.txt
file2.txt  linux_practice  projects_backup
pyogesh2424@cloudshell:~$ ln -s data.txt softlink.txt
pyogesh2424@cloudshell:~$ ls -l
total 16
-rw-rw-r-- 2 pyogesh2424 pyogesh2424 0 May 18 09:38 data.txt
-rw-rw-r-- 1 pyogesh2424 pyogesh2424 0 May 18 09:35 file2.txt
-rw-rw-r-- 2 pyogesh2424 pyogesh2424 0 May 18 09:38 hardlink.txt
drwxrwxr-x 2 pyogesh2424 pyogesh2424 4096 May 18 09:30 linux_practice
drwxrwxr-x 2 pyogesh2424 pyogesh2424 4096 May 18 09:40 projects
drwxrwxr-x 2 pyogesh2424 pyogesh2424 4096 May 18 09:39 projects_backup
-rwxr-xr-x 1 pyogesh2424 pyogesh2424 913 May 18 08:54 README-cloudshell.txt
lrwxrwxrwx 1 pyogesh2424 pyogesh2424 8 May 18 09:40 softlink.txt -> data.txt
pyogesh2424@cloudshell:~$
```

```
pyogesh2424@cloudshell:~$ mkdir file_practice
cd file_practice
pyogesh2424@cloudshell:~/file_practice$ pwd
/home/pyogesh2424/file_practice
pyogesh2424@cloudshell:~/file_practice$ echo "Linux Commands Practice" > notes.txt
echo "Google Cloud Shell" >> notes.txt
echo "DevOps and MLOps" >> notes.txt
echo "Learning Linux Basics" >> notes.txt
echo "Practice Makes Perfect" >> notes.txt
pyogesh2424@cloudshell:~/file_practice$ ls
notes.txt
pyogesh2424@cloudshell:~/file_practice$ cat notes.txt
Linux Commands Practice
Google Cloud Shell
DevOps and MLOps
Learning Linux Basics
Practice Makes Perfect
pyogesh2424@cloudshell:~/file_practice$ echo "Second File" > file2.txt
pyogesh2424@cloudshell:~/file_practice$ cat notes.txt file2.txt
Linux Commands Practice
Google Cloud Shell
DevOps and MLOps
Learning Linux Basics
Practice Makes Perfect
Second File
pyogesh2424@cloudshell:~/file_practice$ more notes.txt
Linux Commands Practice
Google Cloud Shell
DevOps and MLOps
Learning Linux Basics
Practice Makes Perfect
pyogesh2424@cloudshell:~/file_practice$ less notes.txt
pyogesh2424@cloudshell:~/file_practice$ tail notes.txt
Linux Commands Practice
Google Cloud Shell
DevOps and MLOps
Learning Linux Basics
Practice Makes Perfect
pyogesh2424@cloudshell:~/file_practice$ head notes.txt
Linux Commands Practice
Google Cloud Shell
DevOps and MLOps
Learning Linux Basics
Practice Makes Perfect
```

Cloud Shell Editor

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Learning Linux Basics
Practice Makes Perfect
pyogesh2424@cloudshell:~/file_practice$ less notes.txt
pyogesh2424@cloudshell:~/file_practice$ tail notes.txt
Linux Commands Practice
Google Cloud Shell
DevOps and MLOps
Learning Linux Basics
Practice Makes Perfect
pyogesh2424@cloudshell:~/file_practice$ head notes.txt
Linux Commands Practice
Google Cloud Shell
DevOps and MLOps
Learning Linux Basics
Practice Makes Perfect
pyogesh2424@cloudshell:~/file_practice$ nano notes.txt
pyogesh2424@cloudshell:~/file_practice$ cat notes.txt
Linux Commands Practice
Google Cloud Shell
DevOps and MLOps
Learning Linux Basics
Practice Makes Perfect
pyogesh2424@cloudshell:~/file_practice$ gedit notes.txt
-bash: gedit: command not found
pyogesh2424@cloudshell:~/file_practice$ awk '{print}' notes.txt
Linux Commands Practice
Google Cloud Shell
DevOps and MLOps
Learning Linux Basics
Practice Makes Perfect
pyogesh2424@cloudshell:~/file_practice$ echo "Yogesh 21" > students.txt
echo "Rahul 22" >> students.txt
echo "Arun 20" >> students.txt
pyogesh2424@cloudshell:~/file_practice$ awk '{print $1}' students.txt
Yogesh
Rahul
Arun
pyogesh2424@cloudshell:~/file_practice$ sed 's/Linux/Unix/' notes.txt
Unix Commands Practice
Google Cloud Shell
DevOps and MLOps
Learning Unix Basics
Practice Makes Perfect
pyogesh2424@cloudshell:~/file_practice$
```



```
Rahul
Arun
pyogesh2424@cloudshell:~/file_practice$ sed 's/Linux/Unix/' notes.txt
Unix Commands Practice
Google Cloud Shell
DevOps and MLOps
Learning Unix Basics
Practice Makes Perfect
pyogesh2424@cloudshell:~/file_practice$ mkdir permission_practice
cd permission_practice
pyogesh2424@cloudshell:~/file_practice/permission_practice$ touch sample.txt
pyogesh2424@cloudshell:~/file_practice/permission_practice$ ls
sample.txt
pyogesh2424@cloudshell:~/file_practice/permission_practice$ ls -l sample.txt
-rw-rw-r-- 1 pyogesh2424 pyogesh2424 0 May 18 10:28 sample.txt
pyogesh2424@cloudshell:~/file_practice/permission_practice$ chown $USER sample.txt
pyogesh2424@cloudshell:~/file_practice/permission_practice$ ls -l sample.txt
-rw-rw-r-- 1 pyogesh2424 pyogesh2424 0 May 18 10:28 sample.txt
pyogesh2424@cloudshell:~/file_practice/permission_practice$ ls -l sample.txt
-rw-rw-r-- 1 pyogesh2424 pyogesh2424 0 May 18 10:28 sample.txt
pyogesh2424@cloudshell:~/file_practice/permission_practice$ chgrp $(id -gn) sample.txt
pyogesh2424@cloudshell:~/file_practice/permission_practice$ ls -l sample.txt
-rw-rw-r-- 1 pyogesh2424 pyogesh2424 0 May 18 10:28 sample.txt
pyogesh2424@cloudshell:~/file_practice/permission_practice$ umask
0002
pyogesh2424@cloudshell:~/file_practice/permission_practice$ stat sample.txt
  File: sample.txt
  Size: 0          Blocks: 0          IO Block: 4096   regular empty file
Device: 8,17      Inode: 131331       Links: 1
Access: (0664/-rw-rw-r--)  Uid: ( 1000/pyogesh2424)   Gid: ( 1000/pyogesh2424)
Access: 2026-05-18 10:28:59.770310577 +0000
Modify: 2026-05-18 10:28:59.770310577 +0000
Change: 2026-05-18 10:30:10.655336298 +0000
 Birth: 2026-05-18 10:28:59.770310577 +0000
pyogesh2424@cloudshell:~/file_practice/permission_practice$ getfacl sample.txt
-bash: getfacl: command not found
pyogesh2424@cloudshell:~/file_practice/permission_practice$ setfacl -m u:$USER:rwX sample.txt
-bash: setfacl: command not found
pyogesh2424@cloudshell:~/file_practice/permission_practice$ lsattr sample.txt
-----e----- sample.txt
pyogesh2424@cloudshell:~/file_practice/permission_practice$ sudo chattr +i sample.txt
pyogesh2424@cloudshell:~/file_practice/permission_practice$ sudo usermod -aG sudo username
usermod: user 'username' does not exist
pyogesh2424@cloudshell:~/file_practice/permission_practice$
```



```
usermod: user 'username' does not exist
pyogesh2424@cloudshell:~/file_practice/permission_practice$ ifconfig
docker0: flags=4099<UP,BROADCAST,MULTICAST> mtu 1460
    inet 172.17.0.1 netmask 255.255.0.0 broadcast 172.17.255.255
    ether 9a:bc:0a:22:bf:4a txqueuelen 0 (Ethernet)
    RX packets 0 bytes 0 (0.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 0 bytes 0 (0.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1460
    inet 10.88.0.4 netmask 255.255.0.0 broadcast 10.88.255.255
    ether de:73:aa:cf:d6:a2 txqueuelen 1000 (Ethernet)
    RX packets 56987 bytes 36795912 (36.7 MB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 47088 bytes 56412829 (56.4 MB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    loop txqueuelen 1000 (Local Loopback)
    RX packets 36321 bytes 100300945 (100.3 MB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 36321 bytes 100300945 (100.3 MB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

pyogesh2424@cloudshell:~/file_practice/permission_practice$ sudo apt update
sudo apt install net-tools -y
Get:1 https://cli.github.com/packages stable InRelease [3,917 B]
Get:2 https://packages.cloud.google.com/apt gcsfuse-noble InRelease [1,227 B]
Get:3 https://packages.cloud.google.com/apt cloud-sdk InRelease [1,620 B]
Hit:4 http://archive.ubuntu.com/ubuntu noble InRelease
Get:5 http://archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Get:6 https://packages.cloud.google.com/apt cloud-sdk/main amd64 Packages [4,716 kB]
Get:7 https://packages.cloud.google.com/apt cloud-sdk/main all Packages [2,006 kB]
Get:8 http://archive.ubuntu.com/ubuntu noble-backports InRelease [126 kB]
Get:9 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 Packages [2,451 kB]
Get:10 http://archive.ubuntu.com/ubuntu noble-updates/universe amd64 Packages [2,147 kB]
Get:11 http://archive.ubuntu.com/ubuntu noble-updates/restricted amd64 Packages [3,901 kB]
Get:12 https://download.docker.com/linux/ubuntu noble InRelease [48.5 kB]
Get:13 https://download.docker.com/linux/ubuntu noble/stable amd64 Packages [64.9 kB]
Get:14 https://apt.postgresql.org/pub/repos/apt noble-pgdg InRelease [180 kB]
Get:15 https://apt.postgresql.org/pub/repos/apt noble-pgdg/main amd64 Packages [778 kB]
Get:16 http://security.ubuntu.com/ubuntu noble-security InRelease [126 kB]
```



```
pyogesh2424@cloudshell:~/file_practice/permission_practice$ ifconfig
docker0: flags=4099<UP,BROADCAST,MULTICAST> mtu 1460
    inet 172.17.0.1 netmask 255.255.0.0 broadcast 172.17.255.255
    ether 9a:bc:0a:22:bf:4a txqueuelen 0 (Ethernet)
    RX packets 0 bytes 0 (0.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 0 bytes 0 (0.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1460
    inet 10.88.0.4 netmask 255.255.0.0 broadcast 10.88.255.255
    ether de:73:aa:cf:d6:a2 txqueuelen 1000 (Ethernet)
    RX packets 58846 bytes 57381571 (57.3 MB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 48395 bytes 56627098 (56.6 MB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    loop txqueuelen 1000 (Local Loopback)
    RX packets 37109 bytes 100377158 (100.3 MB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 37109 bytes 100377158 (100.3 MB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

pyogesh2424@cloudshell:~/file_practice/permission_practice$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
2: eth0@if8: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1460 qdisc noqueue state UP group default qlen 1000
    link/ether de:73:aa:cf:d6:a2 brd ff:ff:ff:ff:ff:ff link-netnsid 0
    inet 10.88.0.4/16 brd 10.88.255.255 scope global eth0
        valid_lft forever preferred_lft forever
3: docker0: <NO-CARRIER,BROADCAST,MULTICAST,UP> mtu 1460 qdisc noqueue state DOWN group default
    link/ether 9a:bc:0a:22:bf:4a brd ff:ff:ff:ff:ff:ff
    inet 172.17.0.1/16 brd 172.17.255.255 scope global docker0
        valid_lft forever preferred_lft forever

pyogesh2424@cloudshell:~/file_practice/permission_practice$ ping google.com
PING google.com (172.253.118.113) 56(84) bytes of data:
64 bytes from sl-in-f113.1e100.net (172.253.118.113): icmp_seq=1 ttl=114 time=0.920 ms
64 bytes from sl-in-f113.1e100.net (172.253.118.113): icmp_seq=2 ttl=114 time=0.595 ms
```



```
pyogesh2424@cloudshell:~/file_practice/permission_practice$ ss
```

Netid	State	Recv-Q	Send-Q	Peer Address:Port	Local Address:Port	Process
u_str	ESTAB	0	0		* 21065	
u_str	ESTAB	0	0	* 21066	* 29824	
u_str	ESTAB	0	0	* 29825	* 28998	
u_str	ESTAB	0	0	* 28999	* 28588	
u_str	ESTAB	0	0	* 28589	* 29895	
u_str	ESTAB	0	0	* 29896	* 62456	
u_str	ESTAB	0	0	* 62455	* 28585	
u_str	ESTAB	0	0	* 28584	/tmp/tmux-1000/default 44218	
u_str	ESTAB	0	0	* 45145	* 29890	
u_str	ESTAB	0	0	* 29889	* 29823	
u_str	ESTAB	0	0	* 29822	* 29827	
u_str	ESTAB	0	0	* 29826	* 29207	
u_str	ESTAB	0	0	* 29208	* 44211	
u_str	ESTAB	0	0	* 44210	* 43731	
u_str	ESTAB	0	0	* 43730	* 29894	
u_str	ESTAB	0	0	* 29893	* 62458	
u_str	ESTAB	0	0	* 62457	* 29206	
u_str	ESTAB	0	0	* 29205	* 28584	
u_str	ESTAB	0	0	* 28585	* 29893	
u_str	ESTAB	0	0	* 29894	* 20765	
u_dgr	ESTAB	0	0	* 20555	* 62457	
u_str	ESTAB	0	0			

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📄 Cloud Shell Editor

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pygosh2424@cloudshell:~/file_practice/permission_practice$ sudo apt install traceroute -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following NEW packages will be installed:
  traceroute
0 upgraded, 1 newly installed, 0 to remove and 38 not upgraded.
Need to get 60.5 kB of archives.
After this operation, 162 kB of additional disk space will be used.
Get:1 http://archive.ubuntu.com/ubuntu noble/universe amd64 traceroute amd64 1:2.1.5-1 [60.5 kB]
Fetched 60.5 kB in 0s (1,550 kB/s)
debconf: delaying package configuration, since apt-utils is not installed
Selecting previously unselected package traceroute.
(Reading database ... 121190 files and directories currently installed.)
Preparing to unpack .../traceroute_1%3a2.1.5-1_amd64.deb ...
Unpacking traceroute (1:2.1.5-1) ...
Setting up traceroute (1:2.1.5-1) ...
update-alternatives: using /usr/bin/traceroute.db to provide /usr/bin/traceroute (traceroute) in auto mode
update-alternatives: warning: skip creation of /usr/share/man/man1/traceroute.1.gz because associated file /usr/share/man/man1/traceroute.db.1.gz (of link group traceroute) doesn't exist
update-alternatives: using /usr/bin/traceroute6.db to provide /usr/bin/traceroute6 (traceroute6) in auto mode
update-alternatives: warning: skip creation of /usr/share/man/man1/traceroute6.1.gz because associated file /usr/share/man/man1/traceroute6.db.1.gz (of link group traceroute6) doesn't exist
update-alternatives: using /usr/bin/lft.db to provide /usr/bin/lft (lft) in auto mode
update-alternatives: warning: skip creation of /usr/share/man/man1/lft.1.gz because associated file /usr/share/man/man1/lft.db.1.gz (of link group lft) doesn't exist
update-alternatives: using /usr/bin/traceproto.db to provide /usr/bin/traceproto (traceproto) in auto mode
update-alternatives: warning: skip creation of /usr/share/man/man1/traceproto.1.gz because associated file /usr/share/man/man1/traceproto.db.1.gz (of link group traceproto) doesn't exist
update-alternatives: using /usr/sbin/tcptraceroute.db to provide /usr/sbin/tcptraceroute (tcptraceroute) in auto mode
update-alternatives: warning: skip creation of /usr/share/man/man8/tcptraceroute.8.gz because associated file /usr/share/man/man8/tcptraceroute.db.8.gz (of link group tcptraceroute) doesn't exist
Processing triggers for man-db (2.12.0-4build2) ...
pygosh2424@cloudshell:~/file_practice/permission_practice$ traceroute google.com
traceroute to google.com (172.217.194.113), 30 hops max, 60 byte packets
 1  10.88.0.1 (10.88.0.1)  0.085 ms  0.015 ms  0.016 ms
 2  * * *
 3  142.251.247.198 (142.251.247.198)  0.679 ms * *
 4  * * *
 5  * * *
 6  * 209.85.246.53 (209.85.246.53)  1.416 ms 108.170.230.103 (108.170.230.103)  1.649 ms
 7  * * *
 8  * * *
 9  * * *
10  * * *
11  * * *
12  * * *
13  * * *
14  * * *
15  * * *
16  * si-in-f113.1e100.net (172.217.194.113)  0.590 ms *
pygosh2424@cloudshell:~/file_practice/permission_practice$ wget https://example.com
--2026-05-18 10:38:52-- https://example.com/
Resolving example.com (example.com)... 172.66.147.243, 104.20.23.154, 2606:4700:10::6814:179a, ...
Connecting to example.com (example.com)|172.66.147.243|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: unspecified [text/html]
Saving to: 'index.html'

index.html      [ <=>          ]    528  --.-KB/s   in 0s

2026-05-18 10:38:52 (87.8 MB/s) - 'index.html' saved [528]

pygosh2424@cloudshell:~/file_practice/permission_practice$ curl https://example.com
<!doctype html><html lang="en"><head><title>Example Domain</title><meta name="viewport" content="width=device-width, initial-scale=1"><style>body{background:#eee;width:60vw;margin:15vh auto;font-family:system-ui,sans-serif}h1{font-size:1.5em}div{opacity:0.8}a:link,a:visited{color:#348}</s
pygosh2424@cloudshell:~/file_practice/permission_practice$ sudo apt install dnsutils -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
dnsutils is already the newest version (1:9.18.39-0ubuntu0.24.04.3).
0 upgraded, 0 newly installed, 0 to remove and 38 not upgraded.
pygosh2424@cloudshell:~/file_practice/permission_practice$ nslookup google.com
```

Cloud Shell

+

https://shell.cloud.google.com/?pli=1&show=ide%2Cterminal

Chat

Cloud Shell Editor

cloudshell X +

Saving to: 'index.html'

index.html [<=>] 628 --.-KB/s in 0s

2026-05-18 10:38:52 (87.8 MB/s) - 'index.html' saved [528]

pyogesh2424@cloudshell:~/file_practice/permission_practice\$ curl https://example.com

<!doctype html><html lang="en"><head><title>Example Domain</title><meta name="viewport" content="width=device-width, initial-scale=1"><style>body{background:#eee;width:60vw;margin:15vh auto;font-family:system-ui,sans-serif}h1{font-size:1.5em}div{opacity:0.8}a:link,a:visited{color:#348}</s

tyl></head><body><div><h1>Example Domain</h1><p>This domain is for use in documentation examples without needing permission. Avoid use in operations.</p><p>Learn more</p></div></body></html>

pyogesh2424@cloudshell:~/file_practice/permission_practice\$ sudo apt install dnsutils -y

Reading package lists... Done

Building dependency tree... Done

Reading state information... Done

dnsutils is already the newest version (1:9.18.39-Ubuntu0.24.04.3).

0 upgraded, 0 newly installed, 0 to remove and 38 not upgraded.

pyogesh2424@cloudshell:~/file_practice/permission_practice\$ nslookup google.com

Server: 169.254.169.254

Address: 169.254.169.254#53

Non-authoritative answer:

Name: google.com

Address: 74.125.68.113

Name: google.com

Address: 74.125.68.102

Name: google.com

Address: 74.125.68.101

Name: google.com

Address: 74.125.68.138

Name: google.com

Address: 74.125.68.100

Name: google.com

Address: 74.125.68.139

Name: google.com

Address: 2404:6800:4003:c05::0a

Name: google.com

Address: 2404:6800:4003:c05::66

Name: google.com

Address: 2404:6800:4003:c05::71

Name: google.com

Address: 2404:6800:4003:c05::65

pyogesh2424@cloudshell:~/file_practice/permission_practice\$ dig google.com

;; <=> DiG 9.18.39-Ubuntu0.24.04.3-Ubuntu <=> google.com

;; global options: +cmd

;; Got answer:

;; ->HEADER<- opcode: QUERY, status: NOERROR, id: 29290

;; flags: qr rd ra; QUERY: 1, ANSWER: 6, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:

;; EDNS: version: 0, flags:; udc: 512

;; QUESTION SECTION:

;google.com. IN A

;; ANSWER SECTION:

google.com. 288 IN A 74.125.68.113

google.com. 288 IN A 74.125.68.102

google.com. 288 IN A 74.125.68.101

google.com. 288 IN A 74.125.68.138

google.com. 288 IN A 74.125.68.100

google.com. 288 IN A 74.125.68.139

;; Query time: 1 msec

;; SERVER: 169.254.169.254#53(169.254.169.254) (UDP)

;; WHEN: Mon May 18 10:39:37 UTC 2026

;; MSG SIZE rcvd: 135

pyogesh2424@cloudshell:~/file_practice/permission_practice\$

Cloud Shell

https://shell.cloud.google.com/?pli=1&show=ide%2Cterminal

Cloud Shell Editor

cloudshell

```
Name: google.com
Address: 74.125.68.100
Name: google.com
Address: 74.125.68.139
Name: google.com
Address: 2404:6800:4003:c05::8a
Name: google.com
Address: 2404:6800:4003:c05::66
Name: google.com
Address: 2404:6800:4003:c05::71
Name: google.com
Address: 2404:6800:4003:c05::65

pycgesh2424@cloudshell:~/file_practice/permission_practice$ dig google.com

; <<> DLG 9.10.35-0ubuntu0.24.04.3-Ubuntu <<> google.com
;; global options: +cmd
;; Got answer:
;; ->HEADER<- opcode: QUERY, status: NOERROR, id: 29290
;; flags: qr rd ra; QUERY: 1, ANSWER: 6, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 512
;; QUESTION SECTION:
;google.com.                IN      A

;; ANSWER SECTION:
google.com.                288     IN      A       74.125.68.113
google.com.                288     IN      A       74.125.68.102
google.com.                288     IN      A       74.125.68.101
google.com.                288     IN      A       74.125.68.138
google.com.                288     IN      A       74.125.68.100
google.com.                288     IN      A       74.125.68.139

;; Query time: 1 msec
;; SERVER: 169.254.169.254#53(169.254.169.254) (UDP)
;; WHEN: Mon May 18 10:39:37 UTC 2026
;; MSG SIZE rcvd: 135

pycgesh2424@cloudshell:~/file_practice/permission_practice$ sleep 300 &
[1] 3091

pycgesh2424@cloudshell:~/file_practice/permission_practice$ ps
  PID TTY          TIME CMD
  696 pts/2    00:00:00 bash
 3091 pts/2    00:00:00 sleep
 3094 pts/2    00:00:00 ps

pycgesh2424@cloudshell:~/file_practice/permission_practice$ pgrep sleep
657
3091

pycgesh2424@cloudshell:~/file_practice/permission_practice$ sleep 300 &
[2] 3102

pycgesh2424@cloudshell:~/file_practice/permission_practice$ sleep 300 &
[3] 3105

pycgesh2424@cloudshell:~/file_practice/permission_practice$ sleep 300 &
sleep 300 &
sleep 300 &
[4] 3108
[5] 3111
[6] 3114

pycgesh2424@cloudshell:~/file_practice/permission_practice$ sleep 300
fg
sleep 300 &
sleep 400 &
jobs
kill %1
bg
pgrep sleep
```

```
Cloud Shell Editor

cloudshell x + v

pygosh2424@cloudshell:~/file_practice/permission_practice$ fg
-bash: fg: current: no such job
pygosh2424@cloudshell:~/file_practice/permission_practice$ sleep 300 &
[1] 3629
pygosh2424@cloudshell:~/file_practice/permission_practice$ sleep 400 &
[2] 3632
pygosh2424@cloudshell:~/file_practice/permission_practice$ jobs
[1]-  Running                  sleep 300 &
[2]+  Running                  sleep 400 &
pygosh2424@cloudshell:~/file_practice/permission_practice$ kill %1
[1]-  Terminated              sleep 300
pygosh2424@cloudshell:~/file_practice/permission_practice$ bg
-bash: bg: job 2 already in background
pygosh2424@cloudshell:~/file_practice/permission_practice$ pgrep sleep
657
3632
pygosh2424@cloudshell:~/file_practice/permission_practice$ df -h
Filesystem      Size  Used Avail Use% Mounted on
overlay          95G   73G   22G   78% /
tmpfs            64M    0    64M    0% /dev
/dev/sda1        95G   73G   22G   78% /root
/dev/disk/by-id/google-home-part1 4.8G  150M   4.4G   4% /home
/dev/root        2.0G  1.3G  715M   64% /usr/lib/modules
shm              64M    0    64M    0% /dev/shm
pygosh2424@cloudshell:~/file_practice/permission_practice$ ^C
pygosh2424@cloudshell:~/file_practice/permission_practice$ df -h
Filesystem      Size  Used Avail Use% Mounted on
overlay          95G   73G   22G   78% /
tmpfs            64M    0    64M    0% /dev
/dev/sda1        95G   73G   22G   78% /root
/dev/disk/by-id/google-home-part1 4.8G  150M   4.4G   4% /home
/dev/root        2.0G  1.3G  715M   64% /usr/lib/modules
shm              64M    0    64M    0% /dev/shm
pygosh2424@cloudshell:~/file_practice/permission_practice$ ^C
pygosh2424@cloudshell:~/file_practice/permission_practice$ du -sh
8.0K
pygosh2424@cloudshell:~/file_practice/permission_practice$ sudo fdisk -l
sudo: fdisk: command not found
pygosh2424@cloudshell:~/file_practice/permission_practice$ dd if=/dev/zero of=mydisk.img bs=1M count=20
20+0 records in
20+0 records out
20971520 bytes (21 MB, 20 MiB) copied, 0.0169336 s, 1.2 GB/s
pygosh2424@cloudshell:~/file_practice/permission_practice$ mkdir mymount
pygosh2424@cloudshell:~/file_practice/permission_practice$ df -h
Filesystem      Size  Used Avail Use% Mounted on
overlay          95G   73G   22G   78% /
tmpfs            64M    0    64M    0% /dev
/dev/sda1        95G   73G   22G   78% /root
/dev/disk/by-id/google-home-part1 4.8G  170M   4.4G   4% /home
/dev/root        2.0G  1.3G  715M   64% /usr/lib/modules
shm              64M    0    64M    0% /dev/shm
pygosh2424@cloudshell:~/file_practice/permission_practice$ sudo umount mymount
umount: mymount: not mounted.
pygosh2424@cloudshell:~/file_practice/permission_practice$ mount | grep mymount
pygosh2424@cloudshell:~/file_practice/permission_practice$ sudo blkid
/dev/sdb1: UUID="b9f0e455-8c03-483b-a0d6-629a47561fb9" BLOCK_SIZE="4096" TYPE="ext4" PARTUUID="5245a676-01"
/dev/dm-0: LABEL="ROOT-A" SEC TYPE="ext2" BLOCK_SIZE="4096" TYPE="ext4"
/dev/sda4: PARTLABEL="KERN-B" PARTUUID="76f23ba4-cdd3-1143-91d2-9a5f7372c111"
/dev/sda2: PARTLABEL="KERN-A" PARTUUID="2762bea5-16e2-5944-b00a-f19370e6ab65"
/dev/sda12: SEC TYPE="msdos" LABEL FATBOOT="EFI-SYSTEM" LABEL="EFI-SYSTEM" UUID="7BE5-F6D0" BLOCK_SIZE="512" TYPE="vfat" PARTLABEL="EFI-SYSTEM" PARTUUID="1b9d2508-4204-1a41-81b8-2ddc3bfe9b40"
/dev/sda9: PARTLABEL="reserved" PARTUUID="9634bb25-70d7-1a45-8b09-3648a7de5fe8"
/dev/sda10: PARTLABEL="reserved" PARTUUID="9f5dc8da-5cc5-f44e-8289-f69f69e2b1d"
/dev/sda7: PARTLABEL="ROOT-C" PARTUUID="9bf2b5b6-84ac-a94a-b087-360efbf48b7"
/dev/sda5: PARTLABEL="ROOT-B" PARTUUID="969395aa-4a21-b54f-a5ae-3d373e0d8005"
/dev/sda3: LABEL="ROOT-A" SEC TYPE="ext2" BLOCK_SIZE="4096" TYPE="ext4" PARTLABEL="ROOT-A" PARTUUID="ca1d3ffc-74bb-3a4c-872e-8923bf79698e"
/dev/sda1: LABEL="STATE" UUID="9f2dbcb8-9534-4465-bea2-481409599ab8" BLOCK_SIZE="4096" TYPE="ext4" PARTLABEL="STATE" PARTUUID="3154d2f0-8b5c-4447-a5e0-0c5002398a44"
/dev/sda11: PARTLABEL="RWF" PARTUUID="bfed8244-2507-ec48-adfc-b7aa9c3bbbae"
/dev/sda8: LABEL="OEM" UUID="7c475393-34ab-4c9b-aebb-a9cea72447d4" BLOCK_SIZE="4096" TYPE="ext4" PARTLABEL="OEM" PARTUUID="d2d9c06d-32cf-1b49-a5e8-e9c15ebb7dc5"
```



```
Welcome to Cloud Shell! Type "help" to get started, or type "gemini" to try prompting with Gemini CLI.
To set your Cloud Platform project in this session use `gcloud config set project [PROJECT_ID]`.
You can view your projects by running `gcloud projects list`.
pyogesh2424@cloudshell:~$ whoami
pyogesh2424
pyogesh2424@cloudshell:~$ sudo useradd testuser
pyogesh2424@cloudshell:~$ sudo userdel testuser
pyogesh2424@cloudshell:~$ sudo useradd -m developer
pyogesh2424@cloudshell:~$ sudo passwd devuser
passwd: user 'devuser' does not exist
pyogesh2424@cloudshell:~$ sudo chage -l devuser
chage: user 'devuser' does not exist in /etc/passwd
pyogesh2424@cloudshell:~$ sudo chage -E 2026-12-31 devuser
chage: user 'devuser' does not exist in /etc/passwd
pyogesh2424@cloudshell:~$ groups
pyogesh2424 adm sudo docker
pyogesh2424@cloudshell:~$ sudo groupadd developers
pyogesh2424@cloudshell:~$ grep developers /etc/group
developers:x:1002:
pyogesh2424@cloudshell:~$ sudo groupadd projectteam
pyogesh2424@cloudshell:~$ su devuser
su: user devuser does not exist or the user entry does not contain all the required fields
pyogesh2424@cloudshell:~$
```